ASSIGNMENT #1

BASIC CYBERSECURITY RISK ASSESSMENT

Submitted By

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Company: BrightLayer Stores

Type of business: Small business with 8 employees

1. IDENTIFY 5 KEY ASSETS

1. **Employee emails**: Employees will use these to communicate, share information and interact with clients and suppliers. They may also be useful for password recovery.
2. **Cloud storage & Invoices**: Invoices and company documents are stored within the Cloud. This contains important financial and company information which needs to be kept secure.
3. **Website**: The website will represent the company so clients can view their products/services. If the website is damaged this can cause reputation issues.
4. **Shared computers**: Employees will use shared computers to complete their everyday tasks. These may hold company documents, saved passwords and active logins.
5. **Public Wi-Fi Router**: Employees will use the router to connect to the internet. This is a key asset as if the network isn't secure then the company's devices and information can be vulnerable to attackers.

2. IDENTIFY 3 POSSIBLE THREATS

1. **Phishing**: Cybercriminals may attempt to scam employees by sending them fake emails/messages from a trusted company/supplier. The aim of these is to get employees to give away their passwords or click on malicious links.
2. **Unauthorised physical access**: Whilst an employee is away from their shared computer an outsider or employee with malicious intent may try to access it. This would allow them to view confidential documents or active accounts.
3. **Cyberattacks from outside**: Cybercriminals may attempt to find vulnerabilities within the company's website/network by scanning for them. They may attempt to exploit outdated software or insecure network devices.

3. IDENTIFY 3 VULNERABILITIES

1. **Poor password security**: Employees may be using weak or repeated passwords. Multi-factor authentication (MFA) may not be enabled on key accounts causing them to be less secure.
2. **Shared computer with poor security**: Shared computers may be using general logins and employees may leave their account logged in. There may be no auto lock feature.
3. **Poor security update**s: There may not be a regular check for any security updates. This means the website, plugins or router firmware may not be up to date and contain known vulnerabilities.

These are potential vulnerabilities however should be checked in the company's systems

4. IDENTIFY RISKS

1. **Risk of compromised email/cloud account:**

   If an employee gets scammed by a phishing attack and uses a weak/repeated passwords with no MFA enabled then the attacker may gain access to their email or cloud storage. This can cause confidential invoices and company information to be exposed. The attacker may also abuse the payment information or send fake messages to suppliers causing financial loss.

2. **Risk of unauthorised access to company information:**

   If an employee doesn't logout of their shared computer then an unauthorised individual may gain access to the active account. They could view, copy, change or delete important company files/invoices. This can lead to information being leaked, lost and may disrupt day to day operations.

3. **Risk of compromised website/network:**

   Known vulnerabilities may be exploited by attackers due to the outdated software on the website or router firmware. This could lead to the website being defaced, unauthorised access, company information being exposed or disruption to the network. This can cause reputation issues within the company.

5. SUGGEST SECURITY IMPROVEMENTS

1. **Enable Multi Factor Authentication (MFA)**: Enable MFA on all employee emails, cloud storage and website admin accounts. Employees should also use strong unique passwords. This will reduce the risk of compromised accounts.

2. **Strong password management:** Use a trusted password manager. Train employees to recognise phishing emails, suspicious links and keep their passwords safe. If they receive any suspicious emails they should report them.

3. **Secure shared computers:** Ensure each employee has a unique login. Enable auto lock on the screen after a short time. Ask employees to logout once they have finished.

4. **Regular security updates:** Ensure all software on the website, plugins, OS and router firmware are up to date. Have a regular patch management system in place to reduce known vulnerabilities.

5. **Secure wi-fi router:** Change the default admin password. Disable any unecessary remote management features and WPS. Ensure the router is using WPA2-AES/WPA3. Keep guest wi-fi separated from company devices.

6. **Keep regular backups:** Keep regular backups of important invoices, company documents and website data. If possible enable versioning on files in the cloud. Regularly test that you can restore from your backup. This will help reduce the impact of accidentally deleting files, corrupting data and cyberattacks.

CONCLUSION

Even though BrightLayer Stores is a small business there are still key cybersecurity risks. Phishing attacks, unauthorised access and outdated software can all effect the company's information, financial information and day to day operations. However by making the suggested security improvements and regularly checking the systems this can reduce the risk.